

The number of trees in a graph

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Abstract

Let T be a tree with t edges. We show that the number of isomorphic (labeled) copies of T in a graph $G = (V, E)$ of minimum degree at least t is at least

$$2|E| \prod_{v \in V} (d(v) - t + 1)^{\frac{(t-1)d(v)}{2|E|}}.$$

Consequently, any n -vertex graph of average degree d and minimum degree at least t contains at least

$$nd(d - t + 1)^{t-1}$$

isomorphic (labeled) copies of T . This answers a question of [3] (where the above statement was proved when T is the path with three edges) while extending an old result of Erdős and Simonovits [4].

1 Introduction

Let T be a t -edge tree and $G = (V, E)$ be a graph with minimum degree at least t . In this note we consider the question of how many (isomorphic) copies of T we can find in G . More precisely, if $V(T) = \{x_1, \dots, x_{t+1}\}$, then we wish to count the number of injections $\phi : V(T) \rightarrow V$ such that $\phi(u)\phi(v) \in E$ for every edge uv of T . This is a basic question in combinatorics, for example, the simple lower bound $\sum_{v \in V} t! \binom{d(v)}{t}$ in the case when T is a star is the main inequality needed for a variety of fundamental problems in extremal graph theory.

A natural way to count walks of length t in a graph G is to add up the entries of A^t , where A is the adjacency matrix of G . The Blakley-Roy [2] inequality uses linear algebra to show that the number of walks of length t is at least nd^t in any graph of average degree d with n vertices (in fact the inequality is a more general statement about inner products). Another approach to counting walks, and more generally homomorphisms of trees, was used by Sidorenko, using an analytic method and the tensor power trick [5]. Erdős and Simonovits [4] proved that in a graph with average degree d , the number of walks of length t that repeat a vertex is a negligible proportion of the total number

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of walks of length t as $d \rightarrow \infty$. Consequently, their result implies that in a graph of average degree d with n vertices there are at least $(1 - o(1))d^t n$ paths with t edges as $d \rightarrow \infty$. On the other hand, in [3] the following lower bound for the number of homomorphic copies of T in a graph $G = (V, E)$ is proved, where a homomorphic copy is a (not necessarily injective) function $\phi : V(T) \rightarrow V$ such that $\phi(u)\phi(v) \in E$ for every edge uv of T :

$$2|E| \prod_{v \in V} d(v)^{\frac{(t-1)d(v)}{2|E|}}. \quad (1)$$

Combining (1) with the result of [4], we obtain that the number of isomorphic copies of a path T in G is at least

$$(1 - o(1))2|E| \prod_{v \in V} d(v)^{\frac{(t-1)d(v)}{2|E|}}.$$

The result of Erdős and Simonovits [4] does not give a precise expression for the $o(1)$ error term above (although presumably it could be worked out from their proof).

In [3] the following more precise lower bound was given in the case when $T = P_3$, the path with three edges, and G has minimum degree at least 3:

$$2|E| \prod_{v \in V} (d(v) - 2)^{\frac{2d(v)}{2|E|}}. \quad (2)$$

The authors in [3] asked whether a bound similar to (1) and (2) could be proved for the number of isomorphic copies of a tree T in G assuming that G has sufficiently large minimum degree. The spirit of the question was to obtain a bound that is a convex function of the degrees of the vertices (and in particular whose unique minimum occurs when G is regular). Here we provide such a bound that generalizes (2).

Theorem 1. *Let T be a tree with t edges and G be an n -vertex graph with average degree d and minimum degree at least t . Then the number of isomorphic (labeled) copies of T in G is at least*

$$nd \prod_{v \in V} (d(v) - t + 1)^{\frac{(t-1)d(v)}{nd}}.$$

A consequence of this is the following lower bound in terms of the average degree in G .

Corollary 2. *Let T be a tree with t edges and G be an n -vertex graph with average degree d and minimum degree at least t . Then the number of isomorphic (labeled) copies of T in G is at least*

$$nd(d - t + 1)^{t-1}.$$

Indeed, Corollary 2 follows immediately from Theorem 1 by applying Jensen's inequality to the function $f(x) = (t-1)x \log(x - t + 1)$ which is convex for $x \geq t$. Note also that the Corollary is nearly sharp as shown by complete graphs. Indeed, if G is the n vertex graph of disjoint cliques, each with $d+1$ vertices and $d \geq t$, then the number of copies of T in G is $nd(d-1) \cdots (d-t+1)$.

The proof of Theorem 1 uses the ideas first introduced by Alon, Hoory and Linial [1], and subse-

quently developed in [3] to count homomorphisms of T in G .

2 Proof of Theorem 1

We start with a graph G of minimum degree at least t and a tree T with t edges. Let Ω be the set of all isomorphic copies of T in G . In other words, Ω is the set of injections $\xi : V(T) \rightarrow V(G)$ such that $\xi(u)\xi(v) \in E(G)$ for every $uv \in E(T)$. Label the vertices of T by first fixing a leaf x_1 and then labeling vertices $x_2, x_3, \dots$ such that for any $j > 1$ there is a unique $f(j) < j$ such that $x_j x_{f(j)} \in E(T)$. We could, for example, label the vertices using Breadth-First Search or Depth-First Search. Let us call such a labeling of T *good*.

We consider *oriented* isomorphisms $\phi : V(T) \rightarrow V(G)$ which can be constructed as follows. Start with an arbitrary (directed) edge $v_1 v_2 \in E(G)$ and map x_1 to v_1 and x_2 to v_2 . Once $x_1, x_2, \dots, x_i \in V(T)$ are embedded as $\omega_1, \omega_2, \dots, \omega_i \in V(G)$, i.e., $\phi(x_j) = \omega_j$ for $j \leq i$, then choose an arbitrary neighbor ω_{i+1} of $\omega_{f(i+1)}$ outside $\{\omega_1, \omega_2, \dots, \omega_i\}$ and embed x_{i+1} as ω_{i+1} . This gives us a natural probability on the sample space Ω of isomorphic copies of T in G , with associated probability measure $\mathbb{P}$. For convenience, given $\omega \in \Omega$, we let ω_i denote the i th vertex of ω in the embedding. This probability measure is defined on a specific isomorphic copy $\omega \in \Omega$ of T in G by

$$\mathbb{P}(\omega) = \frac{1}{nd} \prod_{i=2}^t \frac{1}{|N(\omega_{f(i+1)}) \setminus \{\omega_1, \omega_2, \dots, \omega_i\}|}.$$

Since $|N(\omega_{f(i+1)}) \setminus \{\omega_1, \omega_2, \dots, \omega_i\}| \geq d(\omega_{f(i+1)}) - t + 1$,

$$\mathbb{P}(\omega) \leq \frac{1}{nd} \prod_{i=2}^t \frac{1}{d(\omega_{f(i+1)}) - t + 1} := p(\omega).$$

Let d be the average degree of G and n the number of vertices. Then, by the inequality of arithmetic and geometric means (using that $\sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1$),

$$|\Omega| \geq \prod_{\omega \in \Omega} \mathbb{P}(\omega)^{-\mathbb{P}(\omega)} \geq \prod_{\omega \in \Omega} p(\omega)^{-p(\omega)} = nd \prod_{\omega \in \Omega} \prod_{i=2}^t (d(\omega_{f(i+1)}) - t + 1)^{p(\omega)}.$$

Interchanging the products we get

$$|\Omega| \geq nd \prod_{i=2}^t \prod_{\omega \in \Omega} (d(\omega_{f(i+1)}) - t + 1)^{p(\omega)}.$$

A term in the product above of the form $d(v) - t + 1$ appears when v is the i th vertex of some $\omega \in \Omega$, for some $i : 2 \leq i \leq t$. Therefore, we have

$$|\Omega| \geq nd \prod_{i=2}^t \prod_{v \in V} (d(v) - t + 1)^{g_i(v)} \tag{3}$$

where

$$g_i(v) := \sum_{\substack{\omega \subset G \\ \omega_i = v}} p(\omega).$$

The key part of the proof is to show

$$g_i(v) \geq \frac{d(v)}{nd}. \quad (4)$$

We note that here is where our proof differs from the previous works [1, 3]. Those papers dealt with homomorphisms instead of isomorphisms, so there was no need to avoid previously embedded vertices of T , and the corresponding probability distribution in that setting is

$$\mathbb{P}'(\omega) = \frac{1}{nd} \prod_{i=2}^t \frac{1}{d(\omega_{f(i+1)})}.$$

Moreover, if we use the probability measure $\mathbb{P}'$ (instead of the function p which is not a probability measure), then (4) actually holds with equality essentially because the Markov chain associated with the distribution $\mathbb{P}'$ is reversible. This is not true in our case, and there are constructions showing that in our situation, $g_i(v) > d(v)/nd$ is possible. Consequently, the argument showing (4) in our situation is more delicate.

To this end, we will prove (4) by proving the following stronger statement by induction on t : Given a t -edge tree T with good labeling $x_1, \dots, x_{t+1}$ and associated function f , an n -vertex graph $G = (V, E)$ with average degree d , and $1 \leq i \leq t+1$, we have $g_i(v) \geq d(v)/nd$. Note that we have included $i = 1$ and $i = t+1$ in this statement as this will be needed in the induction argument that we will use.

The case $t = 1$ is trivial (for both $i = 1$ and $i = 2$) so assume that $t > 1$. Let us first assume that $i < t+1$. Let $T' = T - x_{t+1}$ be the tree obtained from T by deleting the leaf x_{t+1} , let $\omega^- = \omega_1, \dots, \omega_t$ and $N = N(\omega_{f(t+1)}) \setminus \{\omega_1, \dots, \omega_t\}$ so that $|N| \geq d(\omega_{f(t+1)}) - t + 1$. Then

$$\begin{aligned} g_i(v) &= \sum_{\omega^-: \omega_i = v} \sum_{\omega_{t+1} \in N} p(\omega) \\ &\geq \sum_{\omega^-: \omega_i = v} (d(\omega_{f(t+1)}) - t + 1) p(\omega) \\ &\geq \sum_{\omega^-: \omega_i = v} \frac{1}{nd} \prod_{i=2}^t \frac{1}{d(\omega_{f(i+1)}) - t + 2}. \end{aligned}$$

Finally, we note that the rightmost expression is precisely $g_i(v)$ for the tree T' which has $t-1$ edges. So by induction it is at least $d(v)/nd$ as required.

We now suppose that $i = t+1$. Given a copy $\omega = \omega_1, \dots, \omega_{t+1}$ of T in G with $\phi(x_i) = \omega_i$ as usual, let us relabel the vertices with $z = z_1, \dots, z_{t+1}$ such that $z_1 = \omega_{t+1}$, $z_{t+1} = \omega_1$, and z is a good labeling of $\omega = \phi(T)$. Note that this is clearly possible as we may just produce a good labeling of

ω beginning with ω_{t+1} and ending with ω_1 (recall that x_1 is a leaf of T). As before, define

$$p(z) := \frac{1}{nd} \prod_{j=2}^t \frac{1}{d(z_{f(j+1)}) - t + 1}.$$

Now we make the crucial observation that $p(\omega) = p(z)$. To see this, observe that

$$p(\omega) = \frac{1}{nd} \prod_{j=2}^t \frac{1}{d(\omega_{f(j+1)}) - t + 1} = \frac{1}{nd} \prod_{j=2}^t \left(\frac{1}{d(\omega_j) - t + 1} \right)^{d_T(x_j) - 1}.$$

as each term is counted once for each child of the corresponding vertex as the good labeling is constructed. As $\{z_2, \dots, z_t\} = \{\omega_2, \dots, \omega_t\}$, we also obtain

$$p(\omega) = \frac{1}{nd} \prod_{j=2}^t \left(\frac{1}{d(\omega_j) - t + 1} \right)^{d_T(x_j) - 1} = \frac{1}{nd} \prod_{j=2}^t \left(\frac{1}{d(z_j) - t + 1} \right)^{d_T(x_{\pi(j)}) - 1} = p(z),$$

where π is the permutation on $t - 1$ elements such that $z_j = x_{\pi(j)}$ for all $2 \leq j \leq t$. Consequently,

$$g_{t+1}(v) = \sum_{\omega: \omega_{t+1}=v} p(\omega) = \sum_{z: z_1=v} p(\omega) = \sum_{z: z_1=v} p(z) = g_1(v) \geq \frac{d(v)}{nd}.$$

Inserting this into (3) we get

$$|\Omega| \geq nd \prod_{v \in V} (d(v) - t + 1)^{\frac{(t-1)d(v)}{nd}}.$$

This proves the theorem. □

3 Concluding Remarks

- If the maximum degree of the subgraph induced by any tree with t edges is k , then the above proof gives a better bound:

Corollary 3. *Fix a tree T with t edges. Let $G = (V, E)$ be an n -vertex graph such that copy of T in G induces a subgraph of maximum degree at most k , and such that G has minimum degree at least k . Then the number of isomorphic copies of T in G is at least*

$$2|E| \prod_{v \in V} (d(v) - k + 1)^{\frac{(t-1)d(v)}{2|E|}}.$$

- We were not able to decide if the following statement is true (even for paths):

Fix a tree T with t edges. The number of isomorphic labeled copies of T in an n -vertex graph of large enough minimum degree and average degree d is at least $nd(d-1) \cdots (d-t+1)$.

This statement if true would be best possible, since a graph consisting of disjoint cliques of order $d + 1$ has average degree d and exactly $nd(d - 1) \cdots (d - t + 1)$ isomorphic copies of any tree with t edges.

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